CS 132 Compiler Construction, Spring 2025

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Single Choice Exam, May 1, 2025

ID

Name

This exam consists of 48 questions. Each question has four options, exactly one of which is correct, while the other three options are incorrect.

I will grade each question in the following way.

- If you check *none* of the options, you get 0 points.
- If you check *one* option, and that option is correct, you get 2 points. If you check *one* option, and that option is wrong, you get –0.667 points (yes, negative!).

The maximum point total is $48 \times 2 = 96$ points. I will calculate a percentage based on the points in the following way:

$$\frac{\max(0, \text{point total})}{96} \times 100$$

Notice that if your point total is negative, you will get 0 percent.
The exam is 90 minutes.

Example
Consider the grammar

$$
\begin{array}{lcl}
A & ::= & A\ z \ \mid\ \epsilon \ \mid\ z\ x\ B \\
B & ::= & z\ A\ y \ \mid\ z\ C \\
C & ::= & \epsilon \ \mid\ C\ A\ x \ \mid\ x\ A
\end{array}
$$

where {A,B,C} is the set of nonterminal symbols, A is the start symbol, {x,y,z} is the set of terminal symbols, and ϵ denotes the empty string.

Question 1
Which nonterminals are nullable?

- a ☐ $\{C\}$
- b ☐ $\{A\}$
- c ☑ $\{A, C\}$
- d ☐ $\{\}$

Question 2
What is First(A)?

- a ☐ $\{x\}$
- b ☐ $\{\}$
- c ☑ $\{z\}$
- d ☐ $\{y, z\}$

Question 3
What is First(B)?

- a ☐ $\{x, z\}$
- b ☐ $\{y, z\}$
- c ☑ $\{z\}$
- d ☐ $\{y\}$

Question 4
What is First(C)?

- a ☐ $\{x, y\}$
- b ☐ $\{z\}$
- c ☑ $\{x, z\}$
- d ☐ $\{y\}$

Question 5
What is Follow(A)?

- a ☐ $\{\}$
- b ☑ $\{\$, x, y, z\}$
- c ☐ $\{\$, y\}$
- d ☐ $\{\$, y, z\}$

Question 6
What is Follow(B)?
- a ☐ $\{y\}$
- b ☐ $\{x, z\}$
- c ☑ $\{\$, x, y, z\}$
- d ☐ $\{z\}$

Question 7
What is Follow(C)?
- a ☐ $\{x, z\}$
- b ☐ $\{\$\}$
- c ☑ $\{\$, x, y, z\}$
- d ☐ $\{x\}$

Question 8
Is the grammar LL(1)?
- a ☐ Yes
- b ☑ No
- c ☐ The question cannot be answered with the information provided.
- d ☐ The LL(1)-checker would go into an infinite loop.

Example
Consider the grammar

$$
\begin{array}{lcl}
A & ::= & C\ z\ B \quad | \quad x\ z\ w \quad | \quad z\ y\ A \\
B & ::= & y\ y \quad | \quad x \quad | \quad w\ B \\
C & ::= & B\ C\ w \quad | \quad z\ B \quad | \quad \epsilon
\end{array}
$$

where {A,B,C} is the set of nonterminal symbols, A is the start symbol, {w,x,y,z} is the set of terminal symbols, and ϵ denotes the empty string.

Question 9
Which nonterminals are nullable?
- a ☐ $\{A\}$
- b ☐ $\{A, C\}$
- c ☑ $\{C\}$
- d ☐ $\{\}$

Question 10
What is First(A)?
- a ☑ $\{w, x, y, z\}$
- b ☐ $\{x, y\}$
- c ☐ $\{\}$
- d ☐ $\{w, x, z\}$

Question 11

What is First(B)?

- a ☐ $\{\}$
- b ☐ $\{w, y\}$
- c ☑ $\{w, x, y\}$
- d ☐ $\{x, z\}$

Question 12

What is First(C)?

- a ☑ $\{w, x, y, z\}$
- b ☐ $\{w, z\}$
- c ☐ $\{w, y, z\}$
- d ☐ $\{\}$

Question 13

What is Follow(A)?

- a ☐ $\{\}$
- b ☑ $\{\$\}$
- c ☐ $\{w, x, y, z\}$
- d ☐ $\{w, x, y\}$

Question 14

What is Follow(B)?

- a ☐ $\{w, x, y\}$
- b ☐ $\{w, x, z\}$
- c ☐ $\{\$, w, y, z\}$
- d ☑ $\{\$, w, x, y, z\}$

Question 15

What is Follow(C)?

- a ☐ $\{\$, w, x, z\}$
- b ☑ $\{w, z\}$
- c ☐ $\{\}$
- d ☐ $\{\$, w, z\}$

Question 16

Is the grammar LL(1)?

- a ☐ Yes
- b ☑ No
- c ☐ The question cannot be answered with the information provided.
- d ☐ The LL(1)-checker would go into an infinite loop.

Example
Consider the grammar

$$
\begin{array}{lcl}
A & ::= & C \;\mid\; B\ x \;\mid\; x\ y\ A \\
B & ::= & z\ y\ w \\
C & ::= & w\ B\ w \;\mid\; \epsilon
\end{array}
$$

where {A,B,C} is the set of nonterminal symbols, A is the start symbol, {w,x,y,z} is the set of terminal symbols, and ϵ denotes the empty string.

Question 17
Which nonterminals are nullable?
- a ☐ $\{A, B\}$
- b ☑ $\{A, C\}$
- c ☐ $\{\}$
- d ☐ $\{A\}$

Question 18
What is First(A)?
- a ☐ $\{w, y\}$
- b ☑ $\{w, x, z\}$
- c ☐ $\{x\}$
- d ☐ $\{\}$

Question 19
What is First(B)?
- a ☐ $\{w, x\}$
- b ☐ $\{\}$
- c ☑ $\{z\}$
- d ☐ $\{y, z\}$

Question 20
What is First(C)?
- a ☐ $\{\}$
- b ☐ $\{w, x, y\}$
- c ☑ $\{w\}$
- d ☐ $\{x\}$

Question 21
What is Follow(A)?
- a ☐ $\{x\}$
- b ☐ $\{\$, x, y, z\}$
- c ☑ $\{\$\}$
- d ☐ $\{w, x, y, z\}$

Question 22

What is Follow(B)?

- a ☐ $\{\$, w, y, z\}$
- b ☑ $\{w, x\}$
- c ☐ $\{\}$
- d ☐ $\{w, y, z\}$

Question 23

What is Follow(C)?

- a ☐ $\{\$, x, z\}$
- b ☐ $\{\$, w\}$
- c ☑ $\{\$\}$
- d ☐ $\{w, x, y, z\}$

Question 24

Is the grammar LL(1)?

- a ☑ Yes
- b ☐ No
- c ☐ The question cannot be answered with the information provided.
- d ☐ The LL(1)-checker would go into an infinite loop.

Example

Consider the grammar

$$
\begin{array}{lcl}
A & ::= & B\ z\ C \ \mid\ x\ w\ w \\
B & ::= & \epsilon \ \mid\ y\ y \\
C & ::= & z\ A \ \mid\ w \ \mid\ \epsilon
\end{array}
$$

where {A,B,C} is the set of nonterminal symbols, A is the start symbol, {w,x,y,z} is the set of terminal symbols, and ϵ denotes the empty string. The grammar is LL(1). The predictive parsing table is a two-dimensional table called *table*.

Question 25

What does table(A,w) contain?

- a ☐ $x\ w\ w$
- b ☐ $z\ A$
- c ☑ error
- d ☐ $B\ z\ C$

Question 26

What does table(A,x) contain?

- a ☑ $x\ w\ w$
- b ☐ w
- c ☐ error
- d ☐ $B\ z\ C$

Question 27

What does table(A,y) contain?

- a ☐ error
- b ☑ $B\ z\ C$
- c ☐ $x\ w\ w$
- d ☐ w

Question 28

What does table(A,z) contain?

- a ☐ $x\ w\ w$
- b ☑ $B\ z\ C$
- c ☐ error
- d ☐ w

Question 29

What does table(B,w) contain?

- a ☑ error
- b ☐ ϵ
- c ☐ $B\ z\ C$
- d ☐ $y\ y$

Question 30

What does table(B,x) contain?

- a ☑ error
- b ☐ $y\ y$
- c ☐ w
- d ☐ ϵ

Question 31

What does table(B,y) contain?

- a ☐ error
- b ☐ $z\ A$
- c ☐ ϵ
- d ☑ $y\ y$

Question 32

What does table(B,z) contain?

- a ☐ w
- b ☐ error
- c ☑ ϵ
- d ☐ $y\ y$

Question 33
What does table(C,w) contain?

- a ☐ ϵ
- b ☐ error
- c ☑ w
- d ☐ $z\ A$

Question 34
What does table(C,x) contain?

- a ☐ $z\ A$
- b ☐ ϵ
- c ☑ error
- d ☐ w

Question 35
What does table(C,y) contain?

- a ☐ $z\ A$
- b ☐ ϵ
- c ☐ w
- d ☑ error

Question 36
What does table(C,z) contain?

- a ☐ error
- b ☑ $z\ A$
- c ☐ ϵ
- d ☐ w

Example
Consider the grammar

$$
\begin{array}{lcl}
A & ::= & z\ C \ \mid\ \epsilon \\
B & ::= & \epsilon \ \mid\ w\ C\ x \\
C & ::= & A \ \mid\ w\ B\ y
\end{array}
$$

where {A,B,C} is the set of nonterminal symbols, A is the start symbol, {w,x,y,z} is the set of terminal symbols, and ϵ denotes the empty string. The grammar is LL(1). Assume that a recursive-descent parser for the above grammer declares a variable `next` of type `token`, and that the program has three procedures `A()`, `B()`, `C()`, and the following main part:

```
void main() { next = getnexttoken(); A(); eat(EOF); }
```

The procedure `getnexttoken()` gets the next token from an input file. The procedure `eat()` is here written in pseudo-code:

```
void eat(token t) { if (t == next) { next = getnexttoken(); } else { error(); }
```

Question 37
The procedure `A()` contains the snippet:

```
if (next == w) { ???? }
```

What is "????"?

- *a* ☐ `/* do nothing */`
- *b* ☐ `eat(w); C(); eat(x);`
- *c* ☑ `error();`
- *d* ☐ `eat(z); C();`

Question 38
The procedure `A()` contains the snippet:

```
if (next == x) { ???? }
```

What is "????"?

- *a* ☐ `eat(w); B(); eat(y);`
- *b* ☐ `eat(z); C();`
- *c* ☑ `/* do nothing */`
- *d* ☐ `error();`

Question 39
The procedure `A()` contains the snippet:

```
if (next == y) { ???? }
```

What is "????"?

- *a* ☐ `/* do nothing */`
- *b* ☑ `error();`
- *c* ☐ `eat(z); C();`
- *d* ☐ `eat(w); C(); eat(x);`

Question 40
The procedure `A()` contains the snippet:

```
if (next == z) { ???? }
```

What is "????"?

- *a* ☑ `eat(z); C();`
- *b* ☐ `/* do nothing */`
- *c* ☐ `eat(w); C(); eat(x);`
- *d* ☐ `error();`

Question 41
The procedure `B()` contains the snippet:

```
if (next == w) { ???? }
```

What is "????"?
- *a* [√] `eat(w); C(); eat(x);`
- *b* [ ] `/* do nothing */`
- *c* [ ] `error();`
- *d* [ ] `A();`

Question 42
The procedure `B()` contains the snippet:

```
if (next == x) { ???? }
```

What is "????"?
- *a* [ ] `/* do nothing */`
- *b* [ ] `eat(w); C(); eat(x);`
- *c* [ ] `A();`
- *d* [√] `error();`

Question 43
The procedure `B()` contains the snippet:

```
if (next == y) { ???? }
```

What is "????"?
- *a* [√] `/* do nothing */`
- *b* [ ] `eat(w); C(); eat(x);`
- *c* [ ] `error();`
- *d* [ ] `A();`

Question 44
The procedure `B()` contains the snippet:

```
if (next == z) { ???? }
```

What is "????"?
- *a* [ ] `/* do nothing */`
- *b* [√] `error();`
- *c* [ ] `eat(w); C(); eat(x);`
- *d* [ ] `eat(w); B(); eat(y);`

Question 45
The procedure `C()` contains the snippet:

```
if (next == w) { ???? }
```

What is "????"?

- a ☐ `A();`
- b ☑ `eat(w); B(); eat(y);`
- c ☐ `error();`
- d ☐ `eat(z); C();`

Question 46
The procedure `C()` contains the snippet:

```
if (next == x) { ???? }
```

What is "????"?

- a ☐ `eat(w); B(); eat(y);`
- b ☐ `error();`
- c ☑ `A();`
- d ☐ `eat(z); C();`

Question 47
The procedure `C()` contains the snippet:

```
if (next == y) { ???? }
```

What is "????"?

- a ☐ `eat(w); B(); eat(y);`
- b ☑ `error();`
- c ☐ `eat(z); C();`
- d ☐ `A();`

Question 48
The procedure `C()` contains the snippet:

```
if (next == z) { ???? }
```

What is "????"?

- a ☐ `eat(w); B(); eat(y);`
- b ☐ `error();`
- c ☑ `A();`
- d ☐ `eat(w); C(); eat(x);`